Fanny Traore

Computer Systems Engineering Co-op Student | Software, Full-Stack Web, Embedded Systems & Testing
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EDUCATION

Carleton University – B.Eng. in Computer Systems Engineering (Co-op), 3rd year — Expected April 2029
Completed Coursework: Foundations of Imperative Programming (C on Linux), Object-Oriented Software Development (Java, UML, JUnit), Algorithms & Data Structures (Python), Data Management (Python), Signals and Systems (MATLAB), ELEC 2501 Circuits and Signals, ELEC 2507 Electronics I, CCDP 2100 Communication Skills for Engineering Students
In Progress (Fall 2026): Operating Systems (Bash), Intro to Real-Time Systems (embedded C on MSP432)

SKILLS

Languages: Python, C, Java, JavaScript, SQL, C++ (Arduino), Verilog, HTML/CSS, MATLAB, Simulink, Bash, PowerShell
Web & Data: React, Node.js, Express, REST APIs, SQLite, Recharts, Vite, Electron, Bootstrap 5, NumPy, pandas
Testing & Tools: Jest, Supertest, Vitest, pytest, JUnit, GitHub Actions (CI), Git, GitHub, Jira, Linux, Vivado, Figma
Hardware & Lab: FPGA (Zynq-7000), Arduino, UART, KiCad, oscilloscope, spectrum analyzer, logic analyzer, multimeter

PROJECTS

GroovyLog: Full-Stack Music Journal | *React, Node.js, Express, SQLite, Jest, CI* — Oct 2026

- **Built a full-stack web app** with a **React** front end and a **Node.js/Express REST API** (7 routes) for logging album listens.
- Built a **two-table SQLite schema** linked by a **foreign key**, with **SQL queries using JOIN, GROUP BY, AVG and HAVING** for a Recharts stats dashboard.
- **Integrated the iTunes Search API** to fetch album covers with a fallback, and added **input validation** returning clear 400 and 404 errors.
- Added **9 API tests (Jest, Supertest)** and **2 React component tests (Vitest)**, run by **GitHub Actions CI** on every push.

ScamFree: Phishing Detection for Gmail (team of 2) | *JavaScript, Python, Jest, pytest, CI* — Jan 2026 – Sept 2026

- **Built the Chrome extension** that sends Gmail content to the team's REST API and **shows non-blocking phishing alerts**.
- Wrote a **Python data-validation module** and 10 of its 11 pytest tests, including a **7,000-email integration test**.
- Set up a Jest suite with a **95% accuracy quality gate** and **GitHub Actions CI** on Linux for every push and pull request.
- **Fixed a data-loader defect** caught by the tests and logged it as a GitHub Issue; when a failing CI run showed the fix never reached main, restored it and **required passing checks before merging**.

Mochadoro: Desktop App + Arduino Hardware | *Electron, JavaScript, C++, UART* — Dec 2025 – Sept 2026

- Achieved **sub-50 ms two-way sync** between an Electron desktop app and an Arduino over UART using a **state machine**.
- **Wrote C++ Arduino firmware** for RGB status LEDs, a 7-segment counter and a buzzer; **built the entire UI** in HTML, CSS and JavaScript.
- **Found why a display segment stayed dark** by checking the wiring against the datasheet pinout and firmware pin IDs: the **display was inserted backwards**.

PYNQ-Z2 Morse Code Decoder | *Verilog, Vivado, FPGA* — Sept 2026 – Present

- Built a **Morse decoder on a Zynq-7020 FPGA** (synchronizer, clock divider, 4-state FSM), **meeting timing closure**.
- **Isolated FSM chatter** found in board testing with LED debug output; **fixed it with pulse conditioning**.

Bilingual Portfolio Website | *HTML5, CSS, Bootstrap 5, WCAG* — Sept 2026

- **Built an English/French site** in HTML5, CSS and Bootstrap 5 on GitHub Pages; **audited it with WAVE and Lighthouse**, fixing contrast and alt text to reach **accessibility 100**.

LEADERSHIP & TEAMS

FPGA Summer Club – Team Leader, Carleton University — May 2026 – Aug 2026

- **Led a three-person team** building a **real-time video filter** on a Zybo Z7 board; authored the XDC pin-constraint file.

Autonomous Robotics Carleton (ARC) – Software and Electrical Team Member — Sept 2026 – Present

Carleton Planetary Robotics (CPRT) – Software Team Member — Sept 2025 – Dec 2025

- **Researched mounting options** after evaluating rover vibration; **proposed elastomer mounts and bracing** for sensor stacks; used Jira for task tracking.